$$\left.\begin{aligned} v_a' &= \frac{v_0}{2}\cos\omega_1 t \\ v_b' &= -\frac{v_0}{2}\cos\omega_1 t \\ v_c' &= \frac{v_0}{2}\cos\omega_2 t \\ v_d' &= -\frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (2)$$

四物块在质心坐标系中的坐标分别为

$$\left.\begin{aligned} X_a' &= -\frac{l}{2} + \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_b' &= \frac{l}{2} - \frac{v_0}{2\omega_1}\sin\omega_1 t \\ X_c' &= \frac{l}{2} + \frac{v_0}{2\omega_2}\sin\omega_2 t \\ X_d' &= \frac{3l}{2} - \frac{v_0}{2\omega_2}\sin\omega_2 t \end{aligned}\right\} \qquad (3)$$

建立相对桌面静止的坐标系，其原点与第一次碰撞刚碰完时 ab 系统的质心 M_1 重合，X 轴与 X' 轴重合，则相对桌面坐标系，四物块的坐标分别为

$$\left.\begin{aligned} X_a &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_1}\sin\omega_1 t - \frac{l}{2} \\ X_b &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_1}\sin\omega_1 t + \frac{l}{2} \\ X_c &= \frac{1}{2}v_0 t + \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{l}{2} \\ X_d &= \frac{1}{2}v_0 t - \frac{v_0}{2\omega_2}\sin\omega_2 t + \frac{3l}{2} \end{aligned}\right\} \qquad (4)$$

相对桌面坐标系，各物块的速度

$$\left.\begin{aligned} v_a &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_1 t \\ v_b &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_1 t \\ v_c &= \frac{v_0}{2} + \frac{v_0}{2}\cos\omega_2 t \\ v_d &= \frac{v_0}{2} - \frac{v_0}{2}\cos\omega_2 t \end{aligned}\right\} \qquad (5)$$

1. $k_1 = k_2$ 时

设两个弹簧的劲度系数都是 k，则有

$$\omega_1 = \omega_2 = \sqrt{\frac{2k}{m}} = \omega_0 \qquad (6)$$

第一次碰撞后 a、b、c、d 相对桌面坐标系的位置和速度可由(4)式与(5)式求得